higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 3

(12010273)

4 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: TIMBER CONSTRUCTION.

1.1 Choose a correct word/item from column A that matches the description in column B. Write only the letter (A–J) next to question number (1.1.1–1.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Sapwood	A	part of a tree that carries sap from the leaves to the cells
1.1.2	Pith	B	dead/old wood at the centre of a tree
1.1.3	Cambium	C	the central core of the trunk
1.1.4	Heartwood	D	it is also called the stem of a tree
1.1.5	Hardwood	E	layer of living cells underneath the bark
1.1.6	Early wood	F	vertical line that passes from the pith to the bark
1.1.7	Latewood	G	tree that bears flowers
1.1.8	Ray	H	growth ring formed during dry season
1.1.9	Inner bark	I	newly formed outer wood
1.1.10	Trunk	J	growth ring formed during wet season

(10 × 1) (10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 Softwood is also called an angiosperm.

1.2.2 Hardwood is a tree that bears cones.

1.2.3 Softwood loses its leaves in winter.

1.2.4 Hardwood trees are evergreen.

1.2.5 Pine is an example of a hardwood.

(5 × 1) (5)

[15]

QUESTION 2: METALS

2.1 Name FIVE general safety rules to follow when cutting and drilling metals. (5)

2.2 Differentiate between alloy and pure metals. (4)

2.3 When you work with metals, you need to know about methods you can use to change the properties of metals.

Explain the process of changing metal properties using the following:

2.3.1 Drawing

2.3.2 Annealing

2.3.3 Work hardening

(3 × 3) (9)

2.4 Define the term *punching.* (2)
[20]

QUESTION 3: STEEL CONSTRUCTION

3.1 Describe the following aspects of the behaviour of steel:

3.1.1 Elastic behaviour

3.1.2 Toughness

3.1.3 Yield point

3.1.4 Young's modulus

3.1.5 Elongation

(5 × 2) (10)

3.2 The process of prefabrication usually involves the construction of the foundations on site while the wall and roof sections are assembled in a factory and transported to the site when required.

Name FIVE reasons for the design of steel structures in prefabricated sections. (5)

3.3 Name FIVE characteristics of cold-formed steel. (5)
[20]

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QUESTION 4: UNDERSTANDING AND COMBATING CORROSION OF METALS

4.1 Name the FIVE criteria that must be checked before a hand-over certificate can be issued. (5)

4.2 Define the term *braced bay*. (2)

4.3 Steel-framed buildings need bracing to stabilise them against sideways swaying and to prevent the parts that make up the roof, such as trusses and girders, from buckling or distorting.

Name FIVE main functions of bracing. (5)

4.4 Name THREE acceptance tolerances for erected buildings in accordance with prescribed national standards. (3)

[15]

QUESTION 5: JOINING METALS BY SOLDERING, BRAZING AND WELDING

5.1 Explain the following methods used for joining metals together:

5.1.1 Welding

5.1.2 Brazing

5.2.3 Soldering

(3 × 2) (6)

5.2 Name FIVE safety gear needed during welding works. (5)

5.3 Write the following abbreviations in full:

5.3.1 SMAW

5.3.2 GMAW

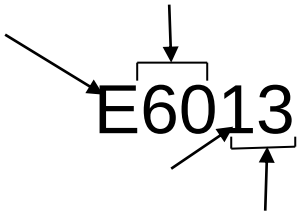
5.3.3 FCAW

5.3.4 TIG

5.3.5 SAW

(5 × 1) (5)

5.4 Interpret the electrode code for a shielded metal arc welding given below:



(4)

[20]

QUESTION 6: PROPERTIES OF GLASS AND THE CUTTING, DRILLING AND SHAPING OF GLASS

6.1 Where will you use the following glass during construction works. Give ONE example of each:

6.1.1 Toughened glass

6.1.2 Laminated glass

6.1.3 Wired glass

6.1.4 Glazed panels

6.1.5 Float glass

(5 × 1) (5)

6.2 Name FIVE advantages of laminated glass. (5)

[10]

TOTAL: 100

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